

Practical: Windows Defender Antivirus Configuration and Threat Detection

Aim

To perform a full system scan using **Windows Defender Antivirus**, enable real-time protection, configure a weekly scheduled scan, simulate a harmless antivirus detection using the **EICAR test file**, and customize Windows Defender notifications.

Requirements

- Windows 10/11 computer or Windows Server with Microsoft Defender Antivirus
 - Administrator access
 - Internet connection for the EICAR test file
 - EICAR test file from the official EICAR website
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Task 1 – Perform a Full System Scan Using Windows Defender

Objective

To perform a complete system scan and verify whether any threats are detected.

Procedure

Step 1: Open Windows Security

1. Click **Start**.
2. Search for **Windows Security**.
3. Open **Windows Security**.

Step 2: Open Virus & Threat Protection

1. Click **Virus & threat protection**.
2. Under **Current threats**, click **Scan options**.

Step 3: Select Full Scan

1. Select **Full scan**.
2. Click **Scan now**.
3. Wait for Windows Defender to complete the scan.

A full scan can take some time depending on the amount of data stored on the computer.

Step 4: Record the Result

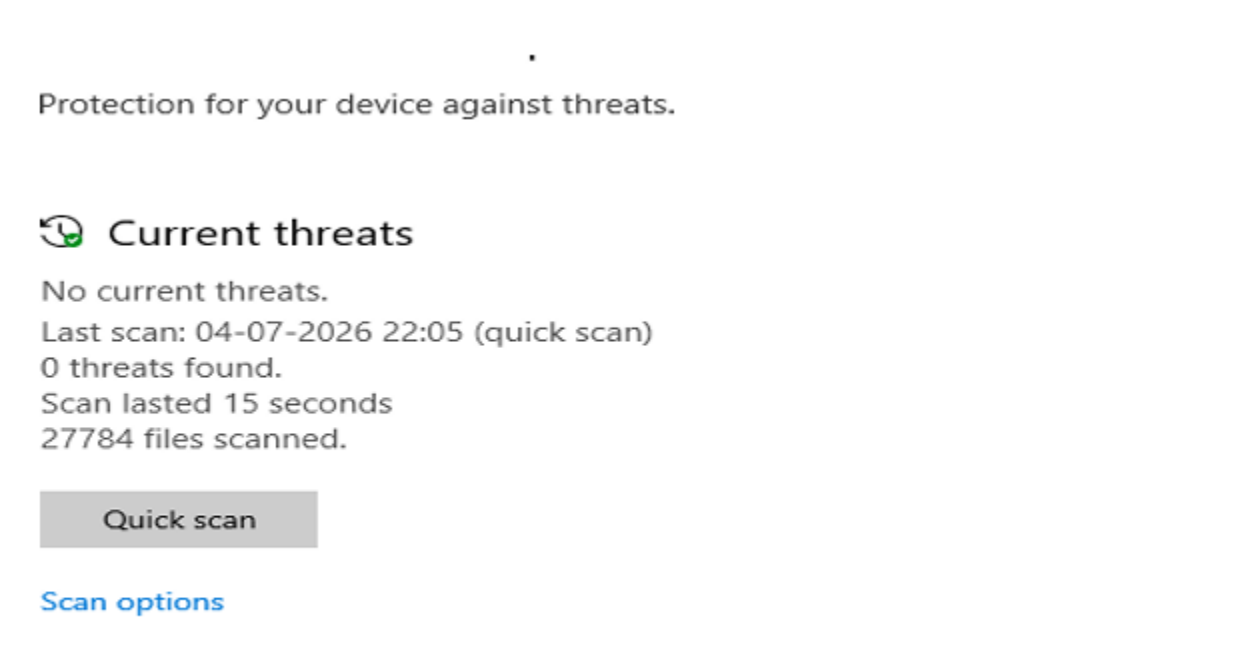
After the scan is completed, record the scan summary.

Sample Result

Item	Result
Scan Type	Full Scan
Status	Scan Completed Successfully
Threats Found	No threats detected
Action Taken	None required

Screenshot

Take a screenshot of the **scan summary** after the scan has completed.



Task 2 – Enable Real-Time Protection and Schedule a Weekly Scan

Objective

To enable Windows Defender real-time protection and configure a scheduled scan to run every Sunday at 8:00 AM.

Part A – Enable Real-Time Protection

Step 1: Open Windows Security

1. Click **Start**.
2. Open **Windows Security**.
3. Select **Virus & threat protection**.

Step 2: Manage Protection Settings

1. Under **Virus & threat protection settings**, click **Manage settings**.
2. Turn **Real-time protection** ON.
3. Make sure **Cloud-delivered protection** is also ON.

Expected Settings

- **Real-time protection:** ON
 - **Cloud-delivered protection:** ON
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Part B – Schedule a Weekly Scan

Step 1: Open Task Scheduler

1. Press **Win + R**.
2. Type:

taskschd.msc

3. Press **Enter**.

Step 2: Open Windows Defender Scheduled Scan

Navigate to:

Task Scheduler Library → Microsoft → Windows → Windows Defender

4. Locate **Windows Defender Scheduled Scan**.
5. Right-click it.

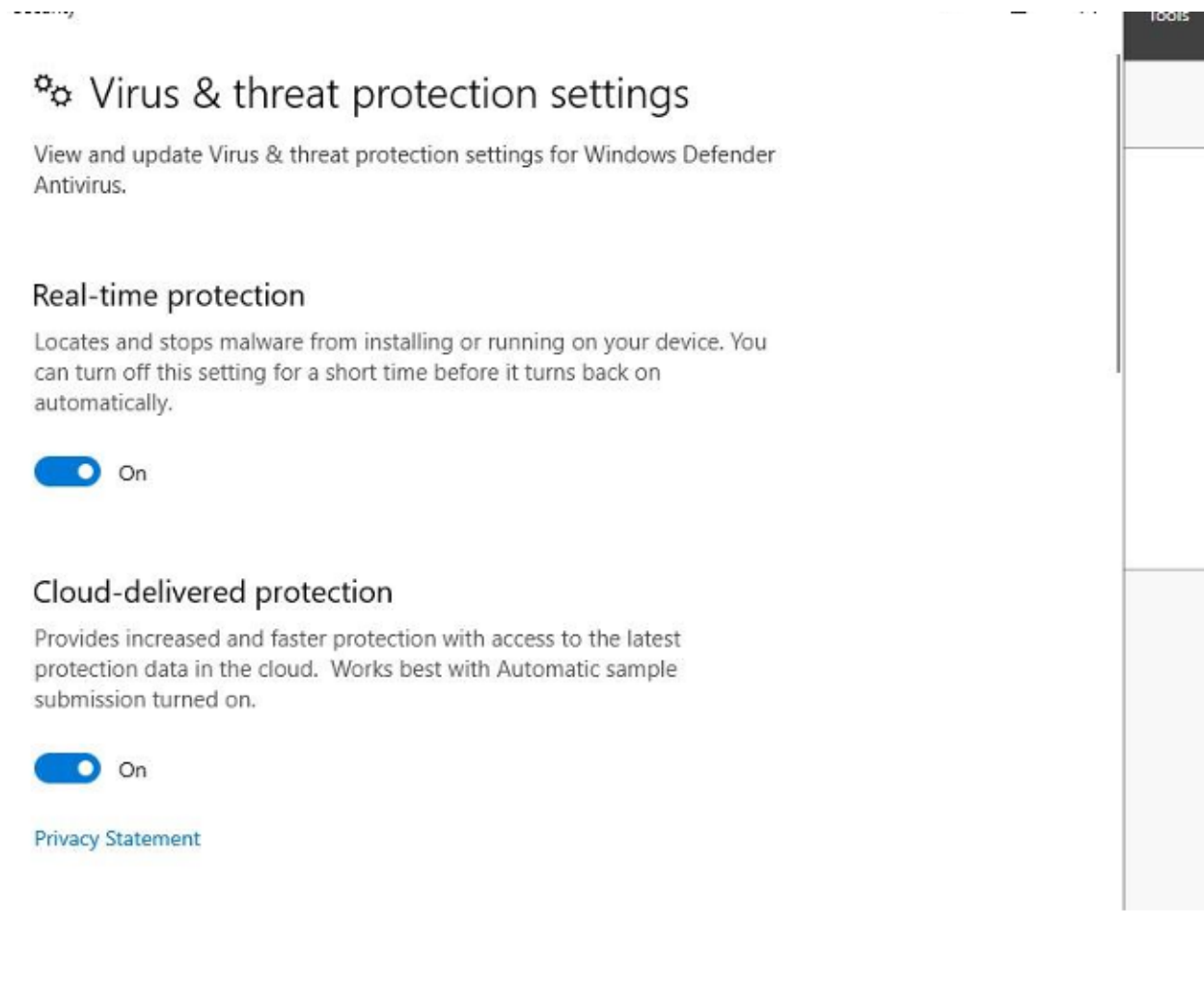
6. Select **Properties**.

Step 3: Configure the Trigger

1. Open the **Triggers** tab.
2. Click **New**.
3. Set:
 - **Begin the task:** On a schedule
 - **Settings:** Weekly
 - **Day:** Sunday
 - **Time:** 8:00 AM
 - **Repeat:** Every 1 week
4. Click **OK**.
5. Click **Apply**.
6. Click **OK**.

Expected Result

Windows Defender is configured to perform the scheduled scan **every Sunday at 8:00 AM**.



Task 3 – Simulate a Harmless Antivirus Threat Using the EICAR Test File

Objective

To test Windows Defender's threat-detection capability using the harmless **EICAR antivirus test file**.

The EICAR test file is an industry-standard test string designed to safely test antivirus software. It is not an actual virus.

Procedure

Step 1: Open the EICAR Website

Open the official EICAR test-file download page:

<https://www.eicar.org/download-anti-malware-testfile/>

Step 2: Download the Test File

1. Download the EICAR test file.
2. Windows Defender should monitor the download in real time.
3. Defender may immediately block, quarantine, or remove the file.

Step 3: Check Windows Security

Open:

Windows Security → Virus & threat protection → Protection history

Look for the EICAR detection.

Expected Result

Windows Defender should identify the test file as an antivirus test threat and take protective action.

Sample Result

Item	Result
Test File	EICAR Test File
Detection	EICAR test threat
Defender Response	Blocked/Quarantined
User Action Required	Remove the test file if it remains

The exact detection name and action may vary depending on the Windows Defender version and security settings.

Task 4 – Configure Windows Defender Notifications

Objective

To customize Windows Defender notifications so that unnecessary notifications are disabled while important threat alerts remain enabled.

Procedure

Step 1: Open Windows Security Settings

1. Open **Windows Security**.

2. Click the **Settings** icon.
3. Select **Manage notifications**.

Step 2: Customize Notifications

1. Review the available Windows Security notification options.
2. Disable unnecessary or informational notifications according to the available settings.
3. Keep important **threat/security notifications** enabled.

The exact notification categories may vary depending on the Windows version.

Step 3: Save the Configuration

Close the settings after making the required changes.

Step 4: Test the Configuration

Use the harmless **EICAR test file** to generate a Defender detection.

1. Download the EICAR test file.
2. Allow Windows Defender to detect and block/quarantine it.
3. Observe the notification generated by Windows Security.
4. Open **Protection history** to verify the detection.

Expected Result

Windows Defender displays an important security notification when the EICAR test file is detected.

Unnecessary informational notifications are reduced according to the configured notification settings.

Final Verification

Perform the following checks after completing the practical:

- Windows Security is opened successfully.
- Virus & threat protection is accessible.

- Full system scan is completed.
 - Scan result is recorded and screenshot is taken.
 - Real-time protection is ON.
 - Cloud-delivered protection is ON.
 - Windows Defender scheduled scan is configured for Sunday at 8:00 AM.
 - EICAR test file is detected by Windows Defender.
 - Defender blocks or quarantines the EICAR test file.
 - Detection appears in Protection History.
 - Windows Defender notifications are customized.
 - Critical security/threat notifications remain enabled.
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After completing a full scan, open the Protection History in Windows Defender and list any threats detected, along with the actions taken (quarantined, removed, allowed). If no threats were found, state 'No threats detected'.

☐ Open Windows Security.

Click Virus & threat protection. Select Protection History.

Review the detected threats and actions taken. Sample Answer (If No Threats Found)

Threat	Action
No threats detected	No action required

Sample Answer (After Testing EICAR)

Threat	Action
EICAR Test File	Quarantined
EICAR Test File	Removed Automatically